MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys). My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.

This work aims to study, from an academical perspective at first, the subject of continuous measurement considering open quantum systems which was a different topic from my supervisor's specific area of research. Our primary motivation was therefore initially to educate ourselves on a field of quantum physics that is well known to a large part of the scientific community. By familiarizing ourselves with the topic, we decided to adopt a particular frame of report due to our journey of understanding. Therefore, the structure of this document is meant to help discovering and exploring the fundamentals of quantum continuous measurement to any unfamiliar reader. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts in the chapter concerning continuously monitored quantum open systems.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2]
However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

Chapter I

Basics of quantum mechanics

I.i Generalities on closed quantum systems

I.i.1 States and time evolution

In quantum mechanics a physical system's state is described by a vector in a Hilbert space:

$$|\psi\rangle \in \mathcal{H}. \tag{I.1}$$

The Hamiltonian $\hat{H}$ is a specific Hermitian[1] operator representing classically the total energy of a system; this relationship will be detailed in Section I.i.2. Physically, the evolution of a system is governed by its energy, the *Schrödinger equation* states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle\,. \tag{I.2}$$

The solution of the Schrödinger equation can be expressed as an unitary[2] operator $\hat{U}$ such that,

$$|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle\,, \tag{I.3}$$

with initial condition $U(t_0,t_0) = \mathbb{I}$. By injecting the previous relation in the state equation, we get

$$\frac{\mathrm{d}\hat{U}}{\mathrm{d}t}(t,t_0) = -\frac{i}{\hbar}\hat{H}(t)\hat{U}(t,t_0). \tag{I.4}$$

We will simply assume the Hamiltonian to be time-independent in the following, nevertheless, solutions exist in the time-dependent case [3]. For us, the equation of evolution, solution of the Schrödinger equation, is

$$|\psi(t)\rangle = \underbrace{e^{-\frac{i\hat{H}(t-t_0)}{\hbar}}}_{\hat{U}(t,t_0)}\,|\psi(t_0)\rangle\,. \tag{I.5}$$

This solution is indeed **unitary** and we can use $\hat{U}^\dagger(t,t_0)$ on the left of both sides of the Eq. (I.5) to obtain $|\psi(t_0)\rangle = \hat{U}^\dagger(t,t_0)\,|\psi(t)\rangle$, i.e., we described **reversible** dynamics.

[1] $\hat{H}^\dagger = \hat{H}$, where † denotes the conjugate transpose, defining the Hermitian adjoint.
[2] $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \mathbb{I}$, where $\mathbb{I}$ is the identity.

I.i.2 Observables

In quantum mechanics, any physical quantity is represented by an **observable** which is an operator fulfilling two properties: the eigenvectors of the observable form an orthonormal basis of the Hilbert space $\mathcal{H}$, and, the corresponding eigenvalues are necessarily real [2]. In finite-dimensional Hilbert spaces, the definition of Hermitian operator[3] is sufficient for it to be an observable. In the case of infinite dimension Hilbert spaces, the definition of an observable requires more rigorous mathematical foundations (self-adjoint operators, the details of which lie beyond the purview of this study). In this sense, we only consider finite dimension cases in the following. We thus define the interval $I_{\mathcal{H}} = [\![1; \dim(\mathcal{H})]\!]$ throughout this document, also, we lighten the notation by omitting $\hat{\bullet}$ symbols on operators hereafter.

A natural operator whose eigenbasis $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ is particularly significant is the Hamiltonian H, as its eigenvalues $\{E_n\}_{n\in I_{\mathcal{H}}}$ correspond to the allowed energy values of a system. We can understand the relation between an operator, its eigenvectors and eigenvalues with its corresponding eigenequation:

$$\forall n \in I_{\mathcal{H}}, \quad H\,|\phi_n\rangle = E_n\,|\phi_n\rangle\,. \tag{I.6}$$

Since $\{|\phi_n\rangle\}_{n\in I_{\mathcal{H}}}$ forms a basis of the Hilbert space, one can describe any state $|\psi\rangle$ in this very basis:

$$\forall\,|\psi\rangle \in \mathcal{H}, \qquad |\psi\rangle = \sum_{n\in I_{\mathcal{H}}} c_n\,|\phi_n\rangle\,, \tag{I.7}$$

where $c_n = \langle\phi_n|\psi\rangle$. This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

Finally, we mention the necessity of normalizing states, which is important when calculating probabilities (the Section I.ii makes it obvious). In order to normalize a state, one needs to set $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$, which is equivalent to the condition $\langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1$.

I.i.3 Purity of states

The concept of *pure state* arises from the necessity to write the state of a system in a different way than that used so far in this document. Indeed, certain ways of preparing a system — such as a statistical mixture[4]— require another formalism because such processes prevent the state of the system from being expressed as a single $|\psi\rangle$ (or as a superposition of state vectors). To take into account such cases, we introduce the density matrix hereafter.

A density matrix ρ is an operator of the Hilbert space $\mathcal{H}$ (and also an operator of the dual space $\mathcal{H}^*$), which follows a set of properties:

- ρ is Hermitian: $\rho = \rho^\dagger$, implying it can be diagonalized and necessarily with a positive eigenvalues, $\mathrm{Sp}(\rho) \subset \mathbb{R}^+$.
- $\mathrm{Tr}\,[\rho] = 1$ necessarily.
- ρ is positive semi-definite: $\forall\,|\psi\rangle \in \mathcal{H}, \quad \langle\psi|\rho|\psi\rangle \geq 0$.

[3]The eigenvectors of a Hermitian operator always span the entire Hilbert space (Spectral Theorem). Using the Gram-Schmidt process, we can always find a complete orthonormal basis from the operator's eigenspaces.

[4]In the Appendix A, we explain how to prepare a system as a statistical mixture or as a superposition of states. However, we recommend to read the Section I.ii on quantum measurement theory before.

The density matrix formalism is particularly useful because it generalizes Dirac formalism. Indeed, if a state can be written as $|\psi\rangle$ then $\rho = |\psi\rangle\langle\psi|$ and we call it a **pure state**. Otherwise, the state is non-pure and can only be expressed with a density matrix. Also, pure states satisfy stronger properties than those defining a general density matrix; specifically, being idempotent $\rho^2 = \rho$ by construction, implying that $\mathrm{Tr}\left[\rho^2\right] = \mathrm{Tr}\left[\rho\right] = 1$ in the case of pure states.

I.i.4 Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathrm{End}(\mathcal{H})$ respecting Properties I.i.3. For pure states we can describe them $|\psi\rangle \in \mathcal{H}$, or, by using the pure matrix density formalism $\rho_p = |\psi\rangle\langle\psi|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.

4. When measuring $\mathcal{A}$ on a normalized state, we know the probability to get one of the possible eigenvalues: $\mathbb{P}[a_n]$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system gives the result a_n, the state of the system immediately after the measurement collapses into the eigenstate $|\phi_n\rangle$.

6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the *Schrödinger* equation:

$$\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = -\frac{i}{\hbar}H(t)\,|\psi(t)\rangle\,. \tag{I.8}$$

   For density matrices, the evolution is driven by the *Liouville-von Neumann* equation:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}\left[H(t), \rho(t)\right]^{1}. \tag{I.9}$$

[1] $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is called the commutator of $\hat{A}$ and $\hat{B}$.

I.ii Quantum Measurement Theory (QMT)

In the QMT [6, 7], measuring consists in applying a projector to a state, so, we call this process a *projective measurement*. The set of projective measurements is called a Projection-Valued Measure (PVM) [8]. For the sake of simplicity, we work — as until now — with non-degenerate spectra for the observables. Since an observable has a set of eigenvectors $\{|a_n\rangle\}_{n\in I_{\mathcal{H}}}$, we can build a projector corresponding to one of them:

$$P_{a_n} = |a_n\rangle\langle a_n| \,. \tag{I.10}$$

This definition implies four important properties:

$$P_{a_n}^\dagger = P_{a_n}, \qquad P_{a_n}^2 = P_{a_n}, \qquad P_{a_n}P_{a_m} = P_{a_n}\delta_{nm}, \qquad \sum_n P_{a_n} = \mathbb{I}.$$

To represent the quantum mechanics definition of a measurement, the projective measurement needs to fulfill some requirements. The first requirement is to generate the probability of observing the result of a measurement on a system. Thus, the probability to observe a_n is defined by

$$\mathbb{P}_{|\psi\rangle}[a_n] = \langle\psi|P_{a_n}|\psi\rangle = |\langle\psi|a_n\rangle|^2, \tag{I.11}$$

$$\mathbb{P}_{\rho}[a_n] = \mathrm{Tr}\left[\rho P_{a_n}\right], \tag{I.12}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

$$\overline{|\psi_{a_n}\rangle} = P_{a_n}\,|\psi\rangle\,, \tag{I.13}$$

$$\overline{\rho_{a_n}} = P_{a_n}\rho P_{a_n}. \tag{I.14}$$

Here the $\overline{\bullet}$ symbol means that the state is not normalized, which will be an important notation for later, see Section II.iii. By normalizing we find the proper expression of a state after a measurement with result a_n:

$$|\psi_{a_n}\rangle = \frac{P_{a_n}\,|\psi\rangle}{\sqrt{\langle\psi|P_{a_n}|\psi\rangle}}, \tag{I.15}$$

$$\rho_{a_n} = \frac{P_{a_n}\rho P_{a_n}}{\mathrm{Tr}\left[\rho P_{a_n}\right]}. \tag{I.16}$$

The obtained state is said **conditioned by the measurement** result a_n, hence the importance of the index a_n on states obtained after measurement. We point out that the normalization factor is exactly the probability to observe the result a_n for the density matrix (or the square root of the probability, when using ket formalism). In the end, projective measurement fulfill the properties of quantum mechanics by construction because they generate probabilities (respecting the fact they need to be positive and their sum is 1), and can also be used to write the expression of collapsed states after measurement.

However, the definition of PVM has been found too restrictive to represent every kind of measurement allowed by quantum mechanics [9, 10]. One can define a more general kind of measurement, that fulfills quantum mechanics needs while relaxing some properties: the orthogonality between each measure elements is not required to generate probabilities.

Therefore, one can build the so-called Positive Operator-Valued Measure (POVM), implicitly defined by the probability to observe the result a_n:

$$\mathbb{P}_\psi[a_n] = \langle\psi|\Pi_{a_n}|\psi\rangle\,, \tag{I.17}$$

$$\mathbb{P}_\rho[a_n] = \mathrm{Tr}\left[\rho\Pi_{a_n}\right], \tag{I.18}$$

where the operator Π_{a_n} is called the *effect* [7] for the result a. Moreover, the set of all effects $\{\Pi_{a_n}\}_{n\in I_\mathcal{H}}$ forms the POVM. As stated, the only properties we force the POVM to respect are:

$$\text{positive semi-definite: } \forall\,|\phi\rangle \in \mathcal{H},\ \langle\phi|\Pi_{a_n}|\phi\rangle \geq 0,\ \text{and,} \qquad \sum_{n\in I_\mathcal{H}} \Pi_{a_n} = \mathbb{I}.$$

These properties imply that POVM generate proper probabilities. Furthermore, the (normalized) states obtained after finding the result a_n are now,

$$|\psi_{a_n}\rangle = \frac{K_{a_n}\,|\psi\rangle}{\sqrt{\langle\psi|\Pi_{a_n}|\psi\rangle}}, \tag{I.19}$$

$$\rho_{a_n} = \frac{K_{a_n}\rho K_{a_n}^\dagger}{\mathrm{Tr}\left[\rho\Pi_{a_n}\right]}, \tag{I.20}$$

where we introduced the operator K_{a_n} called a *Kraus operator*; they respect the relation $\Pi_{a_n} = K_{a_n}^\dagger K_{a_n}$.

Also, POVM, just like PVM, allow one to describe the process of quantum measurement while *ignoring the results*. As in the thought experiment for statistical mixture of states in the Appendix A, we can describe statistically the state without knowing the measurement results:

$$\rho = \sum_{n\in I_\mathcal{H}} \mathbb{P}[a_n]\rho_{a_n} = \sum_{n\in I_\mathcal{H}} K_{a_n}\rho K_{a_n}^\dagger. \tag{I.21}$$

The corresponding relation for PVM is obtained by replacing both K_{a_n} and $K_{a_n}^\dagger$ by P_{a_n}.

Finally, one can observe that POVM generalize PVM in the sense that projectors are special cases of POVM elements (adding self-adjoint property and orthogonality to POVM gives a PVM). Furthermore, PVMs make states collapse on eigenstates that are necessarily orthogonal to each other, which is not necessarily the case for POVMs. A brief example of POVM forcing a state to collapse onto "non-orthogonal states" is when considering initially the state $|\psi\rangle = \frac{|g\rangle+|e\rangle}{\sqrt{2}} \equiv |+\rangle$. If we choose the Kraus operators $K_0 = |g\rangle\langle g|$ and $K_1 = |+\rangle\langle e|$, which fulfill $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$, and, that form the POVM $\{|g\rangle\langle g|\,, |e\rangle\langle e|\}$. Then, when the result of the measurement is 0, we find that the state collapses to $|\psi_0\rangle = |g\rangle$, and when finding 1 it collapses to $|\psi_1\rangle = |+\rangle$. However, $\langle\psi_0|\psi_1\rangle = \frac{1}{\sqrt{2}} \neq 0$, thus the states on which the initial state can collapse are not orthogonal to each other.

I.iii Open Quantum Systems (OQS)

In the previous part, we saw that one can fully describe a system, observables, states and their evolution. However, this description applies to the ideal case where the quantum system is considered isolated from any external interaction. Instead, in the OQS theory systems interact locally with an environment, which itself interacts with a wider environment, etc... To avoid considering infinite chains of interaction, the theory truncates the environment to a local vicinity of the system. Thus, we approximate the total system T as a closed quantum system, within which the quantum system of interest S is called an open quantum system. Moreover, as explained S couples with its environment E, thus giving rise to a dynamical regime distinct from that of closed quantum systems.

I.iii.1 Building the formalism

We will now follow a certain approach [5] in the derivation of the Lindblad master equation, using the *input-output formalism* [11]. It is however possible to use the *collisional model* instead [12, 13], or deriving the Lindblad master equation with the microscopical approach [3]. Since a full derivation of the input-output formalism is mathematically challenging, we will only provide its key equations and their physical meaning; further details on the input-output theory are available in the Appendix D.

As a consequence of S coupling with its environment E, the Hilbert space of study is divided into $\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E$. We consider as initial state a factorized state: $\rho_T = \rho_S \otimes \rho_E$. From now on we will use a typographic convention where $\rho_T \equiv \boldsymbol{\varrho}$ and $\rho_S \equiv \rho$, such that the last assumption rewrites $\boldsymbol{\varrho} = \rho \otimes \rho_E$. The system Hamiltonian H_S depends on which physical system is studied and is thus left free. Instead, the environment is chosen as a collection of quantum harmonic oscillators, the environment Hamiltonian being thus defined by

$$H_E = \int_0^{+\infty} \hbar\omega b_\omega^\dagger b_\omega \mathrm{d}\omega, \tag{I.22}$$

where the operators of the environment satisfy $\left[b_\omega, b_{\omega'}^\dagger\right] = \delta(\omega - \omega')$. This Hamiltonian represents a continuum of bosonic modes of frequency $\omega \in \mathbb{R}^+$, where each mode is represented by a quantum harmonic oscillator. We define the so-called interaction Hamiltonian by

$$H_I = i\hbar \int_0^{+\infty} \sqrt{\frac{k(\omega)}{2\pi}} \left(c b_\omega^\dagger - c^\dagger b_\omega\right) \mathrm{d}\omega, \tag{I.23}$$

where c is an operator of the system, and $\omega \mapsto k(\omega)$ is a function representing the coupling strength between S and E for each considered mode at frequency ω. This coupling is treated as a perturbation: the interaction has a smaller impact on the dynamics than the free evolution. The expression of H_I used in this work can be found, for example, in the case of a two-level system after using the rotating wave approximation. Consequently by using Eq. (I.22,I.23), we define the total Hamiltonian:

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{I.24}$$

By switching to the interaction picture (detailed in the Appendix B), we can get rid of the free dynamics governed by H_0. Indeed, in the interaction picture the Liouville-von Neumann of Eq. (I.9) becomes

$$\frac{\mathrm{d}\tilde{\boldsymbol{\varrho}}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I, \tilde{\boldsymbol{\varrho}}\right], \tag{I.25}$$

where the expression of the interaction Hamiltonian in the interaction picture $\tilde{H}_I$ will be shown below. Beforehand calculating the expression of $\tilde{H}_I$, we need to make some assumptions:

- the *first Markovian assumption*: we consider the interaction strength to be constant in a (large) bandwidth $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$ (ω_0 is the natural frequency of S), so that $\forall k \in \mathcal{W}, \quad k(\omega) \equiv k$. The coupling strength is considered negligible outside of $\mathcal{W}$;
- the system operator acquires a time-dependent phase in the interaction picture: $\tilde{c}(t) = ce^{-i\omega_0(t-t_0)}$. This kind of assumption is an ansatz, we assume the expression one can find for simple systems in the interaction picture [5];
- we assume $k \ll \omega_0$, moreover, on the timescale $\tau = \frac{1}{k}$ all other time-dependencies are negligible.

We can then express the Hamiltonian of interaction in the interaction picture by

$$\tilde{H}_I(t) = i\hbar\sqrt{k}\Big[\tilde{c}(t)b_t^\dagger - \tilde{c}(t)^\dagger b_t\Big], \tag{I.26}$$

in which we defined the input field operator b_t as

$$b_t = \frac{1}{\sqrt{2\pi}}\int_{\mathcal{W}} b_\omega e^{-i(\omega-\omega_0)(t-t_0)}\mathrm{d}\omega. \tag{I.27}$$

In order to understand the environment operators b_t, we should compute its canonical commutation relation:

$$\begin{aligned}\Big[b_t, b_{t'}^\dagger\Big] &= \frac{1}{\sqrt{2\pi}}\int_{-\theta}^{\theta} e^{-i\omega(t-t')}\mathrm{d}\omega \\ &= \delta(t-t').\end{aligned} \tag{I.28}$$

To obtain the Eq. (I.28), we assumed the frequency bandwidth $\mathcal{W}$ to be larger than all other frequencies of the dynamics, in particular, $k \ll \theta$. This implies that we can integrate over all $\mathbb{R}$ and recover a definition of the dirac distribution. Note that by defining $\mathcal{W} = [\omega_0 - \theta; \omega_0 + \theta]$, we implicitly state that $\omega_0 \geq \theta$ (the $\omega_0 - \theta$ frequency is indeed positive). This condition gives again the third assumption mentioned earlier: $k \ll \theta \leq \omega_0 \Rightarrow k \ll \omega_0$. The commutator we just obtained in the Eq. (I.28) is not satisfying because it is a distribution, not a well-defined quantity. We will thus open a brief parenthesis on input operators, in order to introduce the concept of Wiener process and define this commutator differently.

An important result of quantum stochastic calculus and input-output theory (see Appendices C and D) is that we can build the following operator, called a quantum Wiener process as

$$W(t, t_0) = \int_{t_0}^{t} b_{t'}\mathrm{d}t'. \tag{I.29}$$

We can further derive a differential quantity using calculus, called a quantum Wiener increment:

$$\mathrm{d}W_t = \lim_{\delta t \to \mathrm{d}t}\left[W(t+\delta t, t_0) - W(t, t_0)\right] = b_t\mathrm{d}t. \tag{I.30}$$

Using the Eq. (I.29), its canonical commutation relation is

$$\Big[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\Big] = \mathrm{d}t. \tag{I.31}$$

However, by calculating directly the same commutator with the right hand side of the Eq. (I.30), we find

$$\Big[\mathrm{d}W_t, \mathrm{d}W_t^\dagger\Big] = \Big[b_t, b_t^\dagger\Big]\mathrm{d}t^2. \tag{I.32}$$

When we combine this result with the Eq. (I.31), we get

$$\left[b_t, b_t^\dagger\right] \mathrm{d}t = \mathbb{I}. \tag{I.33}$$

If we now define a set of bosonic operators B_t such that $B_t = b_t\sqrt{\mathrm{d}t}$, then

$$\left[B_t, B_t^\dagger\right] = \mathbb{I}, \tag{I.34}$$

which is the canonical commutation relation of the creation/annihilation operators[5]. Because $\{B_t\}_{t\in\mathbb{R}}$ forms a set of proper annihilation operators of quantum harmonic oscillators, we can formally describe the environnement as a quantum harmonic oscillator (labeled by t), whose canonical operators are B_t and $B_t^\dagger$. In other words, we can formally represent the environnement as a quantum harmonic oscillator with the second quantization using these operators at all time.

We now make use of two more assumptions to derive the evolution of the state $\rho(t)$ of the open quantum system S:

- the *Born-Markov approximation*: $\forall t \in \mathbb{R}$, the total state $\boldsymbol{\varrho}(t)$ is a factorized state with the "same" reduced state for the environment, that is to say

  $$\tilde{\boldsymbol{\varrho}}(t) = \tilde{\rho}(t) \otimes \nu_t, \tag{I.35}$$

  where $\nu_t = |0\rangle\langle 0|_t$ is the vacuum state of the bosonic operator B_t corresponding to time t. Thus, $\forall t \in \mathbb{R}$, $\quad B_t\,|0\rangle_t = 0$ and $B_t^\dagger\,|0\rangle_t = |1\rangle_t$; nothing implies we could keep the exact same vacuum state for each time t, we define ν_t so that it is **the vacuum state** from the basis of the environment **at time t**. This implies

  $$\left\langle \left\{b_t, b_{t'}^\dagger\right\} \right\rangle = \delta(t-t') \overset{Eq.\ (I.28)}{=} [b_t, b_{t'}], \tag{I.36}$$

  which are the Markovian assumptions in this framework [5]. The physical meaning is that the environment subsystems interacting at different times are completely **uncorrelated** (characteristic of a white-noise) and the interaction is instantaneous.

- The global state evolves in an infinitesimal time $\mathrm{d}t$ as

  $$\tilde{\boldsymbol{\varrho}}(t+\mathrm{d}t) = U(t+\mathrm{d}t, t)\,\overbrace{[\tilde{\rho}(t) \otimes \nu_t]}^{\tilde{\boldsymbol{\varrho}}(t)}\,U^\dagger(t+\mathrm{d}t, t), \tag{I.37}$$

  where $U(t+\mathrm{d}t, t) = e^{-\frac{i}{\hbar}\tilde{H}_I(t)dt}$.

[5]The annihilation operator acts on the Fock basis (for one mode) as $B_t\,|n\rangle = \sqrt{n}\,|n-1\rangle$, with the condition $B_t\,|0\rangle = 0$, whereas the adjoint (creation operator) acts as $B_t\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$. Together defining the number operator $\hat{n} = B_t^\dagger B_t$ for which Fock states form its eigenbasis: $\hat{n}\,|n\rangle = n\,|n\rangle$.

I.iii.2 The Lindblad master equation

At this point, we presented enough formalism to calculate the evolution of the state $\boldsymbol{\varrho}$ of the total system in an infinitesimal time dt. However, we are not actually interested in the evolution of the total system, we are studying S. The OQS theory states that we can "trace out" the environment by applying a partial trace[6]

$$\tilde{\rho}(t+dt) = \mathrm{Tr}_E[\tilde{\boldsymbol{\varrho}}(t+\mathrm{d}t)]. \tag{I.38}$$

As explained in the last section, we can use the annihilation/creation operators $B_t/B_t^\dagger$ on the environment state. By injecting results concerning these operators to the interaction Hamiltonian in the interaction picture, Eq. (I.26), the expression of the unitary evolution operator becomes

$$U(t+\mathrm{d}t,t) = \exp\left[\sqrt{k\mathrm{d}t}\Big(\tilde{c}(t)\otimes B_t^\dagger - \tilde{c}^\dagger(t)\otimes B_t\Big)\right]. \tag{I.39}$$

The subtlety here is that the unitary evolution operator is depending on $\sqrt{\mathrm{d}t}$ in particular. So we should expand it to the second order (thus up to dt terms), obtaining

$$\begin{aligned} U(t+\mathrm{d}t,t) \approx \mathbb{I}\otimes\mathbb{I} + \Big(\tilde{c}\otimes B_t^\dagger - \tilde{c}^\dagger\otimes B_t\Big)\sqrt{k\mathrm{d}t} + \\ \Big(\tilde{c}^{\dagger 2}\otimes B_t^2 + \tilde{c}^2\otimes B_t^{\dagger 2} - \tilde{c}\tilde{c}^\dagger\otimes B_t^\dagger B_t - \tilde{c}^\dagger\tilde{c}\otimes B_t B_t^\dagger\Big)\frac{k\mathrm{d}t}{2}. \end{aligned} \tag{I.40}$$

Note that we dropped the time-dependency $\tilde{c}(t)$ notation for the more convenient $\tilde{c}$. We can rewrite the evolution of $\tilde{\boldsymbol{\varrho}}$ in the infinitesimal time dt, Eq. (I.38), using the previous expansion:

$$\begin{aligned} \tilde{\boldsymbol{\varrho}}(t+\mathrm{d}t) &= \tilde{\rho}(t)\otimes|0\rangle\langle 0|_t \\ &+ \Big[\tilde{\rho}(t)\tilde{c}^\dagger\otimes|0\rangle\langle 1|_t + \mathrm{h.c.}\Big]\sqrt{k\mathrm{d}t} \\ &+ \Big\{\sqrt{2}\Big[\tilde{\rho}(t)\tilde{c}^{\dagger 2}\otimes|0\rangle\langle 2|_t + \mathrm{h.c.}\Big] - \Big[\tilde{\rho}(t)\tilde{c}\tilde{c}^\dagger\otimes|0\rangle\langle 0|_t + \mathrm{h.c.}\Big]\Big\}\frac{k\mathrm{d}t}{2} \\ &+ \Big[\tilde{c}\tilde{\rho}(t)\tilde{c}^\dagger\otimes|1\rangle\langle 1|_t\Big]k\mathrm{d}t. \end{aligned} \tag{I.41}$$

By tracing out the environment, the evolution of the system S is

$$\tilde{\rho}(t+\mathrm{d}t) = \tilde{\rho}(t) + k\left[\tilde{c}\tilde{\rho}(t)\tilde{c}^\dagger - \frac{\tilde{c}^\dagger\tilde{c}\tilde{\rho}(t) + \tilde{\rho}(t)\tilde{c}^\dagger\tilde{c}}{2}\right]\mathrm{d}t.$$

Let us rewrite this equation directly as a proper differential equation:

$$\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}t}(t) = k\mathcal{D}[\tilde{c}]\tilde{\rho}(t), \tag{I.42}$$

where, we used the super-operator $\mathcal{D}[\hat{O}]\bullet \equiv \hat{O}\bullet\hat{O}^\dagger - \frac{\{\hat{O}^\dagger\hat{O},\bullet\}}{2}$[7]. All in all, by coming back to the Schrödinger representation, we find the so-called Lindblad master equation stating that

$$\boxed{\frac{\mathrm{d}\rho}{\mathrm{d}t}(t) = -\frac{i}{\hbar}[H_0,\rho(t)] + k\mathcal{D}[c]\rho(t)}. \tag{I.43}$$

[6]The partial trace is mathematically the operation satisfying $\rho_A = \sum_k \langle b_k|\rho_{AB}|b_k\rangle$, where $\rho_{AB}\in\mathcal{H}_{AB}$ and $\{|b_k\rangle\}_k$ is an orthonormal basis of $\mathcal{H}_B$. We thus have $\rho = \mathrm{Tr}_E[\boldsymbol{\varrho}]\in\mathcal{H}_S$.

[7]We define the anticommutator between two operators A and B as $\{A,B\} = AB + BA$.

Chapter II

Continuously measured open quantum systems

In the previous chapters, we guided the discussion in order to find the expression of the Lindblad master equation using the input-output formalism. The environment has been traced out to recover information on the system S only, with a certain impact of its with the environment in the dissipator $\mathcal{D}$. Now, we will continuously monitor the environment by measuring the quantum oscillator state of the environment, expressed as B_t. Also, two kind of measurements will be presented below, namely the photodetection and the homodyne detection. In each case, the reader will be able to find graphical illustrations accompanying the physical equations. These illustrations have been generated in `Julia` using the packages `QuantumToolbox.jl` to simulate dynamics, and `Plots.jl` to generate graphs.

Before jumping into the explanations of each detection process, let us consider the steps from the last chapter (OQS theory) where we did not trace out the environment yet. That is to say, everything until Eq. (I.38). Let $\{|j\rangle\langle j|\}_j$ be a PVM (QMT I.ii), whose projector $|j\rangle\langle j|$ is *conditioning* the total system by the measurement result j. The expression of the conditional state of the total system T at time $t + \mathrm{d}t$ is thus

$$\rho_j^{(c)}(t + \mathrm{d}t) = \frac{|j\rangle\langle j|\, \varrho\, |j\rangle\langle j|}{} \tag{II.1}$$

II.i Photodetection

II.i.1 Deriving the Stochastic Master Equation (SME)

II.ii Homodyne detection

II.iii Linear quantum trajectories

Appendix A

Superposition of states VS statistical mixture

In order to illustrate the concepts of superposition of states and statistical mixture, let us consider to observables A, B such that

$$\forall i \in \{1,2\}, \quad A\left|\phi_i\right\rangle = a_i\left|\phi_i\right\rangle,$$
$$\forall i \in \{1,2\}, \quad B\left|\psi_i\right\rangle = b_i\left|\psi_i\right\rangle.$$

We can define a state $|\Psi\rangle$ in superposition of the eigenstates of the observable B:

$$\boxed{|\Psi\rangle = \lambda_1\left|\psi_1\right\rangle + \lambda_2\left|\psi_2\right\rangle}, \tag{A. 1}$$

where $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ such that $|\lambda_1|^2 + |\lambda_2|^2 = 1$. We thus observe the result b_i with following probability

$$\forall i \in \{1,2\}, \quad \mathbb{P}_\Psi[b_i] = |\langle\psi_i|\Psi\rangle|^2 = |\lambda_i|^2 \equiv p_i. \tag{A. 2}$$

If now we measure the system in order to find one of the eigenvalues of A, a_1 for example, we find

$$\begin{aligned}\mathbb{P}_\Psi[a_1] &= |\langle\phi_1|\Psi\rangle|^2 \\ &= |\lambda_1\langle\phi_1|\psi_1\rangle + \lambda_2\langle\phi_1|\psi_2\rangle|^2.\end{aligned} \tag{A. 3}$$

Moreover, by observing that $\forall i \in \{1,2\}, \quad \mathbb{P}_{\psi_i}[a_1] = |\langle\phi_1|\psi_i\rangle|^2$, we finally find that

$$\boxed{\mathbb{P}_\Psi[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1] + 2\,\mathrm{Re}\{\lambda_1\lambda_2^*\langle\phi_1|\psi_1\rangle\langle\phi_1|\psi_2\rangle^*\}}. \tag{A. 4}$$

We will now prepare a state with the same probabilities $\mathbb{P}[b_i] \equiv p_i$ but which is completely different.

A statistical mixture can be well understood by the following thought experiment:

- We want to prepare N states in total using the eigenstates $\{\psi_1, \psi_2\}$ of B.
- We prepare n_1 states in $|\psi_1\rangle$.
- We prepare n_2 states in $|\psi_2\rangle$, so $n_1 + n_2 = N$.

At this point, we have a set of N systems that we can use. However, if we pick a state randomly the probability to pick $|\psi_i\rangle$ depends on the fraction of it $p_i = \frac{n_i}{N}$. This fraction p_i coincides

theoretically with the previous probability $P_\Psi[b_i]$ for the superposition of states: it represents the probability to find the state $|\psi_i\rangle$ from the general state. If we now use the observable A to measure the system we picked, the probability to find the result a_1 becomes

$$\mathbb{P}_{\text{sm}}[a_1] = \frac{n_1\mathbb{P}_{\psi_1}[a_1] + n_2\mathbb{P}_{\psi_2}[a_1]}{N} = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1],$$

by using the law of total probability. Therefore,

$$\boxed{\mathbb{P}_{\text{sm}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1]}. \tag{A. 5}$$

By comparing the last two boxed equations, one can realize that the mathematical expression for the same probability is written differently depending on if we use a superposition of states or a statistical mixture. However, the probability of the statistical mixture is present in the expression of probability of the superposition of states. This difference of formula for the same probability therefore proves that the Dirac formalism $|\psi\rangle$ cannot represent the kind of system whose probability comes from a statistical mixture. It is in fact impossible to represent a statistical mixture with a pure state. Instead, we need to use the density matrix

$$\rho_{\text{sm}} = p_1 \left|\psi_1\right\rangle\!\left\langle\psi_1\right| + p_2 \left|\psi_2\right\rangle\!\left\langle\psi_2\right|, \tag{A. 6}$$

so that,

$$\mathbb{P}_{\rho_{\text{sm}}}[a_1] = p_1\mathbb{P}_{\psi_1}[a_1] + p_2\mathbb{P}_{\psi_2}[a_1], \tag{A. 7}$$

because by definition $\mathbb{P}_{\rho_{\text{sm}}}[a_1] = \text{Tr}\left[\rho_{\text{sm}} \left|\phi_1\right\rangle\!\left\langle\phi_1\right|\right]$.

Appendix B

Representations of quantum mechanics

Brief complement of quantum mechanics

An average is defined in quantum mechanics by

$$\begin{aligned}\langle O\rangle_{\psi} &= \langle\psi|O|\psi\rangle\,,\\ \langle O\rangle_{\rho} &= \mathrm{Tr}\left[\rho O\right],\end{aligned}$$

where the subscripts $|\psi\rangle$ and ρ refer respectively to the representation of pure states or to density matrix more general formalism.

B.i Schrödinger and Heisenberg pictures

In the Schrödinger picture we saw that $|\psi(t)\rangle = U(t)\,|\psi\rangle$, where $U(t) = e^{-\frac{i}{\hbar}Ht}$ in the case of a time-independent Hamiltonian with $t_0 = 0$. Nevertheless, the following results can be derived in the general case of a time dependent Hamiltonian, [3]. Considering the average value of an observable $A_S(t)$ in the Schrödinger picture (in this picture, the time-dependence of an observable can only be explicit), we find

$$\begin{aligned}\langle A(t)\rangle &= \langle\psi(t)|A_S(t)|\psi(t)\rangle\\ &= \langle\psi(t_0)|U^{\dagger}(t,t_0)A_S(t)U(t,t_0)|\psi(t_0)\rangle &\text{(B. 1)}\\ &= \langle\psi|A_H(t)|\psi\rangle_H\,. &\text{(B. 2)}\end{aligned}$$

This calculation naturally reveals the Heisenberg representation of observables and states. Indeed, we have that $|\psi(t_0)\rangle_S \equiv |\psi\rangle_H$ which links both pictures at the initial time t_0. Furthermore, we can link both representation at all time by calculating the evolution of an observable in the Heisenberg picture:

$$\boxed{\frac{\mathrm{d}A_H}{\mathrm{d}t} = \frac{i}{\hbar}\left[H_H, A_H\right] + \frac{\partial A_S}{\partial t}}. \tag{B. 3}$$

This equation leads to the famous *Ehrenfest theorem* analogous in quantum mechanics:

$$\boxed{\frac{\mathrm{d}\langle A\rangle}{\mathrm{d}t} = \frac{i}{\hbar}\langle[H(t), A(t)]\rangle + \left\langle\frac{\partial A}{\partial t}\right\rangle}. \tag{B. 4}$$

We observe that if the observable doesn't depend explicitly on time, then the dynamics of the operator is fully described in the Heisenberg picture.

B.ii Interaction picture

As mentioned in the Section I.iii, we can represent the coupling between S and E with an interaction Hamiltonian such that

$$H_T = \overbrace{H_S \otimes \mathbb{I}_E + \mathbb{I}_S \otimes H_E}^{H_0} + H_I. \tag{B. 1}$$

Knowing that $|\psi(t)\rangle_S = U(t,t_0)\,|\psi\rangle_S$, we can write that

$$\rho_S(t) = U(t,t_0)\rho_S(t_0)U^\dagger(t,t_0). \tag{B. 2}$$

The evolution operator U is the solution of the Schrödinger equation for the total system T. We know that the free system (dynamically governed by H_0) also follows unitary dynamics, which is described by U_0. Thus, we can define the interaction unitary operator U_I as

$$U_I(t,t_0) = U_0^\dagger(t,t_0)U(t,t_0) \tag{B. 3}$$

$$\Leftrightarrow U(t,t_0) = U_0(t,t_0)U_I(t,t_0) \tag{B. 4}$$

If we now try to calculate the average of an operator (in the total space $\mathcal{H}_T$) using the Schrödinger picture, we can find a new interesting relation by injecting the Eq. (B. 4) in the Eq. (B. 2) and then into the following average:

$$\begin{aligned}\langle O(t)\rangle &= \mathrm{Tr}\left[\rho_S(t)O_S(t)\right] \\ &= \mathrm{Tr}\left[\underbrace{U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0)}_{\tilde{\rho}(t)}\ \underbrace{U_0^\dagger(t,t_0)O_S(t)U_0(t,t_0)}_{\tilde{O}(t)}\right].\end{aligned}$$

We can thus define in the interaction picture:

$$\tilde{\rho}(t) = U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0), \tag{B. 5}$$

$$\tilde{O}(t) = U_0^\dagger(t,t_0)O_S(t)U_0(t,t_0). \tag{B. 6}$$

We can now reverse the Eq. (B. 2) and then use the Eq. (B. 4) to get

$$\rho_S(t_0) = U^\dagger(t,t_0)\rho_S(t)U(t,t_0) = U_I^\dagger(t,t_0)U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0)U_I(t,t_0),$$

and, by injecting this into the Eq. (B. 5), we get

$$\tilde{\rho}(t) = U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0). \tag{B. 7}$$

This result is obvious because a density matrix is also an operator and thus respects the Eq. (B. 6). In the end, we find the results used in the main part of this document:

$$\boxed{\tilde{\rho}(t) = U_I(t,t_0)\rho_S(t_0)U_I^\dagger(t,t_0) = U_0^\dagger(t,t_0)\rho_S(t)U_0(t,t_0)}. \tag{B. 8}$$

By working on the Liouville-von Neumann equation, one can show that by jumping into the interaction picture the dynamics do not depend anymore on the free Hamiltonian H_0:

$$\left(\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\frac{i}{\hbar}[H_T,\rho]\right) \Leftrightarrow \left(\frac{\mathrm{d}\tilde{\rho}}{\mathrm{d}t} = -\frac{i}{\hbar}\left[\tilde{H}_I,\tilde{\rho}\right]\right), \tag{B. 9}$$

where $\tilde{H}_I = U_0^\dagger(t,t_0)H_I U_0(t,t_0)$ as defined. It is thus usual to work in this representation to get rid of free dynamics terms, and then jump back to the Schrödinger picture to recover the proper dynamics of the system.

Appendix C

Stochastic calculus for physics

blabla

Appendix D

A taste of the input-output formalism

blablabla

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